

Android Developer Machine Test

1. Create a new Android application using Kotlin.
2. Follow MVVM Architecture with a clean and scalable project structure.
3. Use Git for version control and maintain proper commit history.
4. Design and implement a Splash Screen.
5. Implement Firebase Authentication using Email/Username and Password.
6. Generate and store the Firebase Cloud Messaging (FCM) token after successful login.
7. Validate user authentication before granting access to the application.
8. Restrict all screens and routes to authenticated users only.
9. Implement session management to keep users logged in until logout.
10. Implement Logout functionality.
11. Create and manage two user roles:
 - Admin
 - User
12. Allow only Admin users to access administrative features.
13. Implement role-based navigation and access control.
14. Display a "Development Mode" status indicator (ON/OFF) for Admin users.
15. Create a Dashboard/Home Screen displaying:
 - Total Contacts Count
 - Total Call Logs Count
 - Total SMS Count
 - Last Sync Date & Time
 - Current User Role
 - Development Mode Status
16. Implement CRUD operations using Firebase Firestore or Firebase Realtime Database:
 - Add Record
 - View Record
 - Update Record
 - Delete Record (with confirmation dialog)
17. Display records using RecyclerView.
18. Implement Search and Filter functionality for all list screens.
19. Request and manage runtime permissions for:
 - Contacts
 - Call Logs
 - SMS
20. Read and display device Contacts.
21. Sync Contacts data to Firebase.

22. Read and display device Call Logs including:

- Name
- Number
- Call Type
- Duration
- Date & Time

23. Sync Call Log data to Firebase.

24. Read and display device SMS messages including:

- Sender
- Message Preview
- Date & Time

25. Sync SMS data to Firebase.

26. Implement manual synchronization using a Sync button.

27. Implement automatic background synchronization.

28. Display synchronization progress and status.

29. Prevent duplicate records during synchronization.

30. Protect all screens using authentication checks.

31. Restrict Admin-only features through role-based permissions.

32. Configure Firebase Security Rules appropriately.

33. Prevent unauthorized access and CRUD operations.

34. Handle the following scenarios:

- No Internet Connection
- Authentication Failure
- Permission Denied
- Firebase Errors
- Empty States
- Synchronization Failures

35. Display appropriate loading indicators during all network and synchronization operations.

36. Deliverables:

- Complete Source Code (Git Repository)
- APK File
- README Documentation

37. README must include:

- Project Setup Instructions
- Firebase Configuration Steps
- Architecture Overview
- Libraries Used
- Screenshots